CYS103 Linear Algebra (2026/09)
TUTORIAL 1

Chapter 1: Linear Equations and Linear Transformations

1.1 System of Linear Equations

1.2 Echelon and Reduced Echelon Forms

1. Solve the following system using elementary row operations on the augmented matrix.

$$2x_1 + 4x_2 = -4$$
$$5x_1 + 7x_2 = 11$$

replacement ✓ scaling ✓ Interchange

$\begin{bmatrix} x_1 & x_2 & \cdots & c \end{bmatrix}$

linear system.

$x_1 = ?$

$x_2 = ?$

$$\begin{matrix} R1 \\ R2 \end{matrix}\begin{bmatrix} 2 & 4 & -4 \\ 5 & 7 & 11 \end{bmatrix} \xrightarrow{R_1 = \frac{1}{2}R_1} \begin{bmatrix} 1 & 2 & -2 \\ 5 & 7 & 11 \end{bmatrix}$$

$$\downarrow R_2 = R_2 + (-5R_1)$$

$$\begin{bmatrix} 1 & 2 & -2 \\ 0 & -3 & 21 \end{bmatrix} \xrightarrow{R_2 = -\frac{1}{3}R_2} \begin{bmatrix} 1 & 2 & -2 \\ 0 & 1 & -7 \end{bmatrix}$$

$$\downarrow R_1 = R_1 + (-2R_2)$$

REF

$$\begin{matrix} & x_1 & x_2 & \\ \end{matrix}\begin{bmatrix} 1 & 0 & 12 \\ 0 & 1 & -7 \end{bmatrix}$$

pivot

Therefore, $(x_1, x_2) = (12, -7)$

or

$x_1 = 12$, $x_2 = -7$. #

2. Solve the following systems.

a.
$$x_2 + 4x_3 = -5$$
$$x_1 + 3x_2 + 5x_3 = -2$$
$$3x_1 + 7x_2 + 7x_3 = 6$$

augmented matrix. → EF. → REF.

$$\begin{bmatrix} 0 & 1 & 4 & -5 \\ 1 & 3 & 5 & -2 \\ 3 & 7 & 7 & 6 \end{bmatrix} \rightarrow \cdots \rightarrow \begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 0 & 2 \end{bmatrix} \rightarrow \cdots \rightarrow \begin{bmatrix} 1 & 0 & -7 & 13 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

(0 = 2)

stop.

Therefore, there is no solution for the linear system. The system is inconsistent. #

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b. $\begin{aligned} x_1 - 3x_2 &= 5 \\ -x_1 + x_2 + 5x_3 &= 2 \\ x_2 + x_3 &= 0 \end{aligned}$

$$\begin{bmatrix} 1 & -3 & 0 & 5 \\ -1 & 1 & 5 & 2 \\ 0 & 1 & 1 & 0 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

pivot; x_1 x_2 x_3 C; → REF.

Therefore, $(x_1, x_2, x_3) = (2, -1, 1)$

or

$x_1 = 2, \quad x_2 = -1, \quad x_3 = 1$

→ exactly one solution

→ system is consistent.

→ unique

*.

3. Determine if the following systems are consistent. (Do not completely solve the systems)

a. $\begin{aligned} x_1 + 3x_3 &= 2 \\ x_2 - 3x_4 &= 3 \\ -2x_2 + 3x_3 + 2x_4 &= 1 \\ 3x_1 + 7x_4 &= -5 \end{aligned}$

$$\begin{bmatrix} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & -2 & 3 & 2 & 1 \\ 3 & 0 & 0 & 7 & -5 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & -2 & 3 & 2 & 1 \\ 0 & 0 & -9 & 7 & -11 \end{bmatrix}$$

$$\downarrow$$

$$\begin{bmatrix} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & 0 & 3 & -4 & 7 \\ 0 & 0 & -9 & 7 & -11 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 3 & 0 & 2 \\ 0 & 1 & 0 & -3 & 3 \\ 0 & 0 & 3 & -4 & 7 \\ 0 & 0 & 0 & -5 & 10 \end{bmatrix}$$

matrix in an echelon form.

leading entries.

∴ The system has an exactly one solution.
Thus, it is consistent. #.

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b. $$\begin{aligned} x_1 \qquad\qquad - 2x_4 &= -3 \\ 2x_2 + 2x_3 \qquad\quad &= 0 \\ x_3 + 3x_4 &= 1 \\ -2x_1 + 3x_2 + 2x_3 + x_4 &= 5 \end{aligned}$$

echelon form

leading entry

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -3 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ -2 & 3 & 2 & 1 & 5 \end{bmatrix} \to \cdots \to \begin{bmatrix} 1 & 0 & 0 & -2 & -3 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The system is consistent because it has the infinitely many solutions (0 = 0). #.

4. Determine which matrices are in reduced echelon form and which are in echelon form.

REF EF

a. $\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ REF.

b. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ It is not an echelon.

c. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ It is not an echelon.

d. $\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ EF.

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5. Row reduced the following augmented matrices into reduced echelon forms.

a. $\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 3 & 9 & 7 & 6 \end{bmatrix}$$

Replacement, Scaling $\downarrow R_2 = -3R_1 + R_2$

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & -5 & -15 \end{bmatrix}$$

$\downarrow$ Scaling, $R_2 = -\frac{1}{5} R_2$

$$\begin{bmatrix} 1 & 3 & 4 & 7 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

Replacement, Scaling $\downarrow R_1 = R_1 + (-4R_2)$

$$\begin{bmatrix} 1 & 3 & 0 & -5 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

b. $\begin{bmatrix} 3 & -4 & 2 & 0 \\ -9 & 12 & -6 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix}$

$$\begin{bmatrix} 3 & -4 & 2 & 0 \\ -9 & 12 & -6 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix}$$

Replacement, Scaling $\downarrow R_2 = 3R_1 + R_2$, $R_3 = 2R_1 + R_3$

$$\begin{bmatrix} 3 & -4 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\downarrow$ Scaling, $R_1 = \frac{1}{3} R_1$

$$\begin{bmatrix} 1 & -\frac{4}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Note :

They system is consistent because if has an infinitely many solutions.